Basics of FE-Analysis
BK10A6400
Lecture 2: Rod/Bar/Truss element

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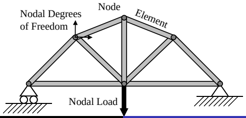
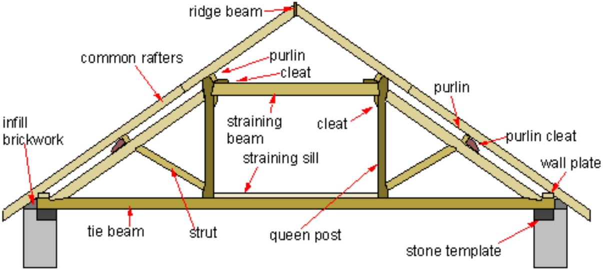
Motivation

- A truss (*ristikko*) is a structure composed of structural elements (such as beams (*palkit*), struts (*tuet*), etc.).
- Joint connections are made by welding or joining the ends with a gusset plate (or nail plates (*naulalevy*) for wooden structures) using rivets (*niitti*) or bolts.
- Truss structures are typically composed of triangles (structural stability).
- Structural members in simple truss structures are two-force members (only under tension or compression, no moments).
- Members are usually straight.
- Simplest truss members (struts) are axially loaded (compression, tension) and some of them serve as so-called zero-force members (stability).



Examples: Traditional roof truss



https://en.wikipedia.org/wiki/Truss/media/File:Queen-post-truss.png

Roof truss with nail plates (modern sauna)

- Members connected with nail plates.
- Plates pressed on both sides of the timber.
- Joints usually modeled as pinned (Eurocode 5, RT 23-10965).
- In reality, joints are semi-rigid (some rotational stiffness).
- Plate manufacturers combine experimental data with Eurocode (EN 1995-1-1) in certified design software.

- Connections via bolts.

Examples: Old bridge of Mansikkakoski (1932-1933), Imatra

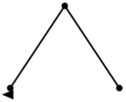
- Connection via gusset plates and rivets.
- Due to this type of connection, structural members also transmit bending moments."
- Complicated to analyze -> simplification.

Examples: Manchester Central Railwaystation : Arched roofs from the 1860s and 1880s.

- Gusset plates and rivets
- Curved (arched) structral members

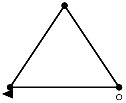
Maxwell's stability criterion (2D trusses, structural mechanics)

(a) Mechanism



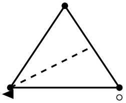
$m = 2,\ j = 3,\ r = 2$
$m + r = 4 < 2j = 6$

(b) Isostatic



$m = 3,\ j = 3,\ r = 3$
$m + r = 6 = 2j = 6$

(c) Redundant



$m = 4,\ j = 3,\ r = 3$
$m + r = 7 > 2j = 6$

m = number of **members** (bars)

j = number of **joints** (nodes)

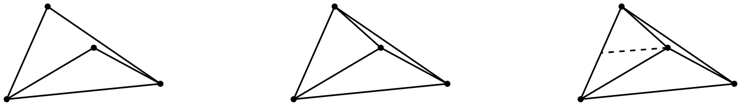
r = number of **support reactions**

Criterion:

$$m + r \begin{cases} < 2j & \text{mechanism (unstable)} \\ = 2j & \text{isostatic (stable)} \\ > 2j & \text{redundant} \end{cases}$$

Maxwell's stability criterion (3D trusses, structural mechanics)

(a) Mechanism **(b) Isostatic** **(c) Redundant**



$m = 5,\ j = 4,\ r = 6$; $m + r = 11 < 3j = 12$

$m = 6,\ j = 4,\ r = 6$; $m + r = 12 = 3j = 12$

$m = 7,\ j = 4,\ r = 6$; $m + r = 13 > 3j = 12$

m = number of **members** (bars)

j = number of **joints** (nodes)

r = number of **support reactions**

Criterion (3D):

$$m + r \begin{cases} < 3j & \text{mechanism (unstable)} \\ = 3j & \text{isostatic (stable)} \\ > 3j & \text{redundant} \end{cases}$$

Two interpretations of Maxwell's stability criterion

Structural mechanics (frames/trusses)

$$m+r \begin{cases} < dj & \text{mechanism (unstable)} \\ = dj & \text{isostatic (stable)} \\ > dj & \text{redundant} \end{cases}$$

- $m =$ members (bars)
- $j =$ joints (nodes)
- $r =$ support reactions
- $d =$ dimension (2 or 3)

$\rightarrow$ *This is what our 2D/3D slides illustrated.*

Architected lattices / metamaterials

$$M = m - dj$$

- $M < 0 \rightarrow$ bending-dominated (mechanism-like, flexible)
- $M \geq 0 \rightarrow$ stretch-dominated (rigid, stiff)

$\rightarrow$ *Used in material science (e.g. Deshpande-Ashby-Fleck). No external supports considered.*

Summary: Same origin, but structural mechanics version includes supports, while the lattice materials version is for

Two interpretations of Maxwell's stability criterion

Structural mechanics (frames/trusses)

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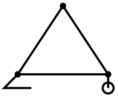

Architected lattices / metamaterials

$$M = m - dj$$

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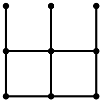
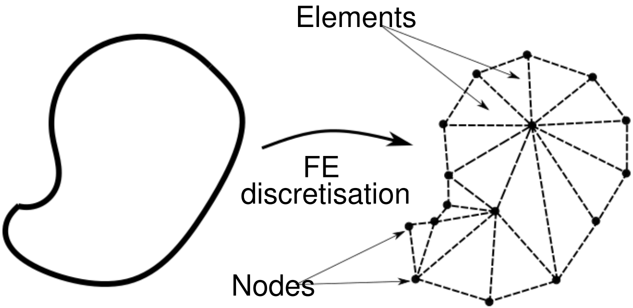


Summary: Same origin, but structural mechanics version includes supports, while the lattice materials version is for infinite/periodic architectures.

Recap: FEM idea

- Differential equations, DEs, (and PDEs) describe problem continuously.
- FEM is a numerical method for PDE.
- In FEM, the original continuous problem (described with DE or PDE) is converted into a linear system of equations by discretizing it.



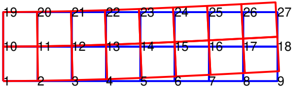
$$\boldsymbol{u} = \boldsymbol{K}^{-1}\boldsymbol{f}_{ext}$$

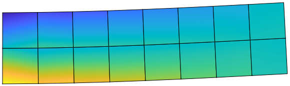
Recap: The procedure of linear displacement-based FEM

1. Formulation of elemental stiffness matrices
   - Derive (or use ready solution) local elemental stiffness matrix $\overline{\boldsymbol{K}}$
2. Assembling global stiffness matrix and a vector of external (global) forces for a structure.
   - Compute global elemental stiffness matrix for a local element k: $\boldsymbol{K}^{(k)} = \boldsymbol{T}^T \overline{\boldsymbol{K}}^{(k)} \boldsymbol{T}$.
   - Assembling global stiffness matrix $\boldsymbol{K}_{sys}$
   - External force vector of a structure $\boldsymbol{f}$
3. Boundary conditions (and global force vector): $\boldsymbol{K}_c$, $\boldsymbol{f}_c$
4. Solving unknown (nodal displacements) from a system of linear equations $\boldsymbol{u}_c = \boldsymbol{K}_c^{-1} \boldsymbol{f}_c$
   - Inverse of stiffness matrix!
   - Solve nodal displacements
5. Solving forces, strains, and stresses. Visualizations.

Recap: Elements and nodes

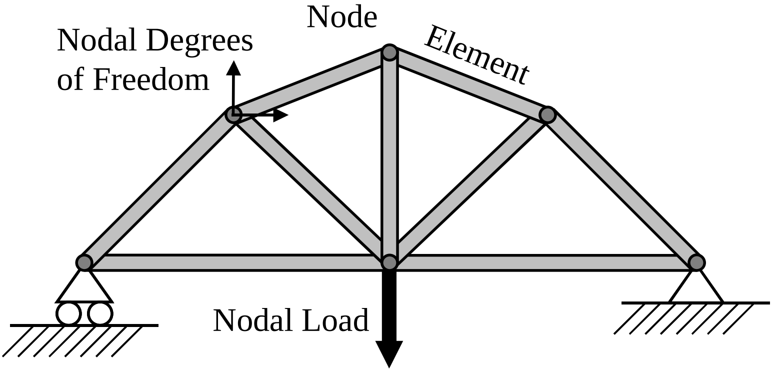
- Degrees of Freedom, element, nodes, nodal displacements and displacement field.
- Elements are interconnected via nodes.
- A hundreds of different element types.
- The most common element types in the structural analysis are rod (truss), beam, shell, and solid elements.
- As an example, a simple cantilever problem is analyzed by four-node plane elements.





Two node rod element

- To analyse truss-type of structures.
- One-dimensional element
- Two nodes, one nodal displacement at each node -> 2 degrees of freedom (DOF)
- Axial loading (elemental level)



Stiffness matrix of a two node rod (truss) element

$$\overline{\boldsymbol{K}} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{1}$$

- From literature (Cook's book p.20, Eq. (2.2.1), Hakala's book p.54, Eq. (2.3.17), for example)
- Where does the stiffness matrix come from?
  - Approximate displacement field with interpolation function
  - Strain (strain-displacement relationship)
  - Stress (stress-strain i.e. constitutive relationship)
  - Energy
  - Stiffness matrix

The principle of minimum potential energy for a rod (truss) element

Displacement field:

$$u_h = \boldsymbol{N}\overline{\boldsymbol{u}} \tag{2}$$

Strain (linear strain - displacement relationship):

$$\varepsilon = \frac{du_h}{dx} = \frac{d\boldsymbol{N}}{dx}\overline{\boldsymbol{u}} = \boldsymbol{B}\overline{\boldsymbol{u}} \tag{3}$$

where $\boldsymbol{B}$ is so-called kinematic matrix.
Constitutive equation (linear stress - strain relationship):

$$\sigma = E\varepsilon \tag{4}$$

Internal and external energies:

$$W_{int} = \frac{1}{2}EA\int_x \varepsilon^2 dx = \frac{1}{2}\overline{\boldsymbol{u}}^T \underbrace{EA\int_0^L \frac{\partial \boldsymbol{N}}{\partial x}^T \frac{\partial \boldsymbol{N}}{\partial x} dx}_{\overline{\boldsymbol{K}}} \overline{\boldsymbol{u}} \tag{5}$$

The principle of minimum potential energy for a rod (truss) element

Internal energy is presented in the form of a stiffness matrix:

$$W_{int} = \frac{1}{2}\overline{\boldsymbol{u}}^T \underbrace{EA \int_0^L \frac{\partial \boldsymbol{N}}{\partial x}^T \frac{\partial \boldsymbol{N}}{\partial x} dx}_{\overline{\boldsymbol{K}}} \overline{\boldsymbol{u}} \tag{6}$$

External energy:

$$W_{ext} = \overline{\boldsymbol{f}}^T \overline{\boldsymbol{u}} \tag{7}$$

Minimizing energy:

$$\frac{\partial \Pi}{\partial \overline{\boldsymbol{u}}} = 0 \tag{8}$$

$$\overline{\boldsymbol{K}}\overline{\boldsymbol{u}} - \overline{\boldsymbol{f}} = 0 \tag{9}$$

General derivation for a stiffness matrix of a rod element

Strain:

$$\varepsilon = \frac{du_h}{dx} \tag{10}$$

Constitutive equation:

$$\sigma = E\varepsilon \tag{11}$$

Strain energy:

$$W_{int} = \frac{1}{2}\int_V \sigma\varepsilon \mathrm{dV} = \frac{1}{2}EA\int_x \varepsilon^2 \mathrm{d}x \tag{12}$$

The vector of elemental internal forces:

$$\overline{\boldsymbol{f}}_{int} = \frac{\partial W_{int}}{\partial \overline{\boldsymbol{u}}} \tag{13}$$

Elemental local stiffness matrix:

$$\overline{\boldsymbol{K}} = \frac{\partial \overline{\boldsymbol{f}}_{int}}{\partial \overline{\boldsymbol{u}}} \tag{14}$$

Displacement field approximation

$$\boldsymbol{u}_h = \boldsymbol{N}\boldsymbol{u} \tag{15}$$

- Shape functions describe shape of the displacement field.
- The shape function gets a value of 1 at that node and a value of zero at all other nodes.
- How to derive shape functions?

Example: Derive shape functions for a two-node rod elements in MATLAB

Displacement field:

$$u_h := a_0 + a_1 x = \boldsymbol{a}^T \boldsymbol{p} \tag{16}$$

Then giving boundary conditions $u_h(x = 0) := u_1$ and $u_h(x = L) := u_2$:

$$\begin{cases} u_h(0) := a_0 = u_1 \\ u_h(L) := a_0 + a_1 L = u_2 \end{cases} . \tag{17}$$

Eqs. (17) in a matrix form:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 1 & L \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}}_{\boldsymbol{a}} = \underbrace{\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}}_{\boldsymbol{u}} . \tag{18}$$

Example: Derive shape functions for a two-node rod element in MATLAB

By solving a_i:s:

$$\boldsymbol{a} = \boldsymbol{A}^{-1}\boldsymbol{u} \tag{19}$$

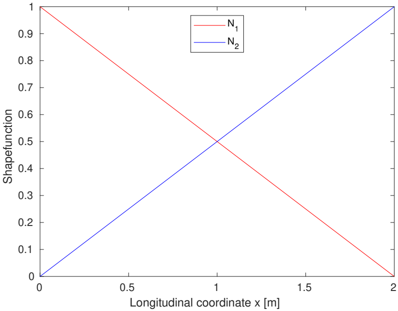
and substituting them into Eq.(16):

$$u_h = (\boldsymbol{A}^{-1}\boldsymbol{u})^T\boldsymbol{p} = \underbrace{\boldsymbol{p}^T\boldsymbol{A}^{-1}}_{\boldsymbol{N}}\boldsymbol{u} \tag{20}$$

where $\boldsymbol{N}$ includes shape functions (known as ansatz functions in German literature).

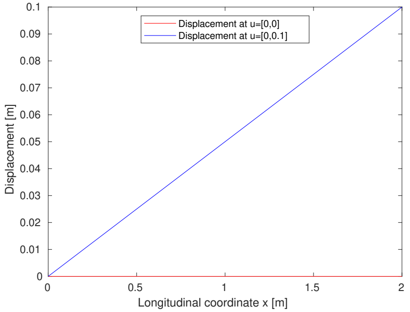
Example: Shape functions

- $N_1 = 1 - x/L$, $N_2 = x/L$
- The shape function gets a value of 1 at that node and a value of zero at all other nodes.
- $N_1(x = 0) = 1$, $N_1(x = L) = 0$
- $N_2(x = 0) = 0$, $N_2(x = L) = 1$



Example: displacement fields

- $u_h(x) = \boldsymbol{N}(x)\boldsymbol{u}$
- $\boldsymbol{N} = [N_1 \quad N_2]$ and $\boldsymbol{u} = [u_1 \quad u_2]^T$



Example: Simple problem with one rod/truss element)

- Element's coordinate system is parallel with a global coordinate system.
- Global nodal displacements (u_X, u_Y) at each node.
- 2 nodes, 1 DOFs at node -> a system of 2 DOF before taking into account the (linear) constraints that are applied eliminating rows and columns of the stiffness matrix for a system.
- Constraints i.e. fixed DOF's: $u_X(x = 0) = 0$

Example: Simple problem with two rod/truss element)

- Element's (local) coordinate system is parallel with a global coordinate system (or vice versa..).
- Global nodal displacements (u_X, u_Y) at each node.
- 3 nodes, 1 DOFs at node -> a system of 3 DOF before taking into account the (linear) constraints that are applied eliminating rows and columns of the stiffness matrix for a system.
- Constraints i.e. fixed DOFâs: $u_X(x = 0) = 0$

Example: Assembling two linear rod elements (a local coordinate system is parallel to a global)

Goes following way if the element's local coordinate system is parallel to the global one, i.e. no need to make a transformation.

$$\overline{\boldsymbol{K}}^{(1)} = \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} \end{pmatrix} \tag{21}$$

$$\overline{\boldsymbol{K}}^{(2)} = \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} \end{pmatrix} \tag{22}$$

A stiffness matrix $\boldsymbol{K}_{sys}$ for a a whole system (a structure of rods)

$$\boldsymbol{K}_{sys} = \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} + k_{11}^{(2)} & k_{12}^{(2)} \\ 0 & k_{21}^{(2)} & k_{22}^{(2)} \end{pmatrix} \tag{23}$$

Example: Assembling three linear rod elements (a local coordinate system is parallel to the global)

Goes following way if the element's local coordinate system is parallel to the global one, i.e. no need to make a transformation.

$$\overline{\boldsymbol{K}}^{(1)} = \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} \end{pmatrix} \tag{24}$$

$$\overline{\boldsymbol{K}}^{(2)} = \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} \end{pmatrix} \tag{25}$$

$$\overline{\boldsymbol{K}}^{(3)} = \begin{pmatrix} k_{11}^{(3)} & k_{12}^{(3)} \\ k_{21}^{(3)} & k_{22}^{(3)} \end{pmatrix} \tag{26}$$

$$\boldsymbol{K}_{sys} = \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} + k_{11}^{(2)} & k_{12}^{(2)} & 0 \\ 0 & k_{21}^{(2)} & k_{22}^{(2)} + k_{11}^{(3)} & k_{12}^{(3)} \\ 0 & 0 & k_{21}^{(3)} & k_{22}^{(3)} \end{pmatrix} \tag{27}$$

Transformation between local and global coordinate systems

- Needs always do if the strain-displacement relationship is approximated (i.e. $\varepsilon \approx \frac{du}{dx}$).
- Elemental stiffnesses have to presented in the structureâs global coordinate system.
- In some cases, transformation results in more degrees of freedom.
- For example, rod element's internal forces f_1 and f_2 are transformed into $f_{G_{X_1}}$,f_{GY_1},f_{GX_2} and f_{GY_2}
- When you compute elemental displacements and member forces, need to transform the global to elemental coordinate systems.

Transformation between local and global coordinate systems

Analyzed structures are not often parallel to the element's longitudinal direction -> Transformation into the global coordinate system.

$$\overline{\boldsymbol{u}} = \boldsymbol{T}\boldsymbol{u}; \quad \boldsymbol{u} = \boldsymbol{T}^T\overline{\boldsymbol{u}} \tag{28}$$

$$\overline{\boldsymbol{f}} = \boldsymbol{T}\boldsymbol{f}; \quad \boldsymbol{f} = \boldsymbol{T}^T\overline{\boldsymbol{f}} \tag{29}$$

$$\boldsymbol{K} = \boldsymbol{T}^T\overline{\boldsymbol{K}}\boldsymbol{T} \tag{30}$$

Transformation for a two node rod element

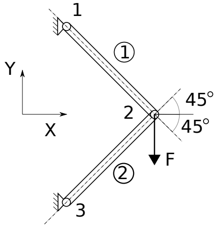
$$
\boldsymbol{T} = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 & 0 \\ 0 & 0 & \cos\alpha & \sin\alpha \end{pmatrix} \tag{31}
$$

$$
\boldsymbol{K} = \boldsymbol{T}^T \overline{\boldsymbol{K}} \boldsymbol{T} \tag{32}
$$

where $\boldsymbol{K}$ is an elemental global stiffness 4×4 matrix for a rod.

Example: Simple truss structure problem

Solve displacements with rod finite elements.



- Global nodal displacements (u_X, u_Y) at each node.
- 3 nodes, 2 DOFs at node -> a system of 6 DOF before taking into account the (linear) constraints that are applied by eliminating rows and columns of the stiffness matrix for a system.
- Constraints i.e. fixed DOF's:
  $u_{X_1} = 0, u_{Y_1} = 0, u_{X_3} = 0, u_{Y_3} = 0$

- Elem 1 consists of nodes 1 and 2
- Elem 2 consists of nodes 2 and 3
- Angle of a rod 1: ?
- Angle of a rod 2: ?

Transformation into a global coordinates system.
Rod 1:

$$\boldsymbol{K}_G^{(1)} = \boldsymbol{T}(\theta_1)^T \overline{\boldsymbol{K}} \boldsymbol{T}(\theta_1) \tag{33}$$

Rod 2:

$$\boldsymbol{K}_G^{(2)} = \boldsymbol{T}(\theta_2)^T \overline{\boldsymbol{K}} \boldsymbol{T}(\theta_2) \tag{34}$$

Example: Simple truss structure problem

Sometimes, it is helpful to present elements' connectivity and corresponding (global) nodal displacements in the table as follows:

	Node I	Node J	nodal displacements
Truss 1	1	2	u_1, v_1, u_2, v_2
Truss 2	2	3	u_2, v_2, u_3, v_3

Note that $u_1, v_1, u_2, v_2, u_3, v_3$ are the structure's global displacement coordinates. Common stiffness components (interconnected via the same node) of global elemental stiffness matrices $\boldsymbol{K}^{(1)}$ and $\boldsymbol{K}^{(2)}$ can be seen from the connectivity table and they are needed to sum to mimic structure correctly. After assembling, the global stiffness matrix should be a size of 6×6.

Elemental stiffness matrices

Rod 1:

$$\boldsymbol{K}_G^{(1)} = \boldsymbol{T}(\theta_1)^T \overline{\boldsymbol{K}} \boldsymbol{T}(\theta_1) \tag{35}$$

$$\boldsymbol{K}_G^{(1)} = \begin{array}{c c} \begin{array}{cccc} u_1 & v_1 & u_2 & v_2 \end{array} & \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} \end{pmatrix} & \begin{array}{c} u_1 \\ v_2 \\ u_2 \\ v_2 \end{array} \end{array}$$

Elemental stiffness matrices

Rod 2:

$$\boldsymbol{K}_G^{(2)} = \boldsymbol{T}(\theta_2)^T \overline{\boldsymbol{K}} \boldsymbol{T}(\theta_2) \tag{36}$$

$$\boldsymbol{K}_G^{(2)} = \begin{array}{c} \begin{matrix} u_2 & v_2 & u_3 & v_3 \end{matrix} \\ \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} \end{array} \begin{matrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{matrix}$$

Assembling

$$\boldsymbol{K}_G = \begin{array}{cc} \begin{array}{cccccc} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 \end{array} & \\ \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & 0 & 0 \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ 0 & 0 & k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ 0 & 0 & k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} & \begin{array}{c} u_1 \\ v_1 \\ u_2 \\ u_2 \\ u_3 \\ v_3 \end{array} \end{array}$$

Applied forces

External forces are set up in a global coordinate system because our global stiffness matrix is also presented in a global coordinate system.
Force F is active at node 2 in a same direction with nodal displacement v_2.

$$\boldsymbol{f}_G = \begin{array}{cc} \begin{pmatrix} 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \end{pmatrix} & \begin{array}{c} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{array} \end{array}$$

Boundary conditions

Fixed nodal displacements are $u_1 = v_1 = u_3 = v_3 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\boldsymbol{f}_{G_{red}} = \begin{array}{cc} \begin{pmatrix} 0 \\ -F \end{pmatrix} & \begin{array}{c} u_2 \\ v_2 \end{array} \end{array}$$

$$\boldsymbol{K}_{G_{red}} = \begin{array}{cc} \begin{array}{cc} \quad u_2 \quad & \quad v_2 \quad \end{array} & \\ \begin{pmatrix} k_{33}^{(1)} + k_{11}^{(2)} & k_{34}^{(1)} + k_{12}^{(2)} \\ k_{43}^{(1)} + k_{21}^{(2)} & k_{44}^{(1)} + k_{22}^{(2)} \end{pmatrix} & \begin{array}{c} u_2 \\ u_2 \end{array} \end{array}$$

Note that $\boldsymbol{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \boldsymbol{K}_{G_{red}} > 0$ (if not, check BC's).
And then displacements (in a global coordinate system) can be solved as

$$\boldsymbol{u}_{G_{red}} = \boldsymbol{K}_{G_{red}}^{-1} \boldsymbol{f}_{G_{red}}$$

Elemental displacements and member forces

After solving $\boldsymbol{u}_{G_{red}}$, local elemental displacements (for an element i) can be compute as follows:

$$\overline{\boldsymbol{u}}^{(i)} = \boldsymbol{T}^{(i)} \boldsymbol{u}_G^{(i)} \tag{37}$$

Member forces (for an element i) can be found as

$$\overline{\boldsymbol{f}}^{(i)} = \overline{\boldsymbol{K}}^{(i)} \overline{\boldsymbol{u}}^{(i)} \tag{38}$$

Note $\boldsymbol{u}_G^{(i)}$ includes element's i (solved nodal coordinates and nodal coordinates at supports).